Technology Stack

SMARTBRIDGE - Technology Stack

- **Date:** 21/09/2026
- **Team ID:** SWTID-2026-7817
- **Team Size:** 4
- **Team Leader:** Karan S
- **Team Members:** SANJAY .M, Santhosh S, Yaseen P
- **Project Name:** AI FitTrack
- **Maximum Marks:** 2 Marks

Technology Stack Details

S.No	Architecture Component / Layer PDF	Technology Chosen PDF	Justification / Purpose PDF
1	**Frontend / Client-Side**	Postman / Thunder Client / React.js	Provides a flexible client interface to execute HTTP REST API calls and render workout JSON payloads.
2	**Backend / Server-Side**	Node.js & Express.js	Lightweight, non-blocking asynchronous event loop allowing fast execution of API endpoints and middleware.
3	**Database / Data Storage**	MongoDB Atlas & Mongoose ODM	Flexible NoSQL document database ideal for dynamic workout schemas and high performance data queries.
4	**Cloud / Hosting / Deployment**	MongoDB Atlas Cloud & Local Node Execution	Provides auto-scaling cloud database hosting with IP whitelisting and reliable connection URIs.

5	**Version Control & CI/CD**	Git & GitHub	Enables seamless team collaboration, code history tracking, and repository management.
6	**Third-Party APIs / Other Tools**	Google Gemini API (@google/genai)	Enables AI workout plan generation (gemini-2.5-flash) and personalized fitness insight analytics.